Preserving Ariel Cylinders and Distance Pieces, Mineral, Synthetic PAO, and Synthetic Ester Oil Compatible

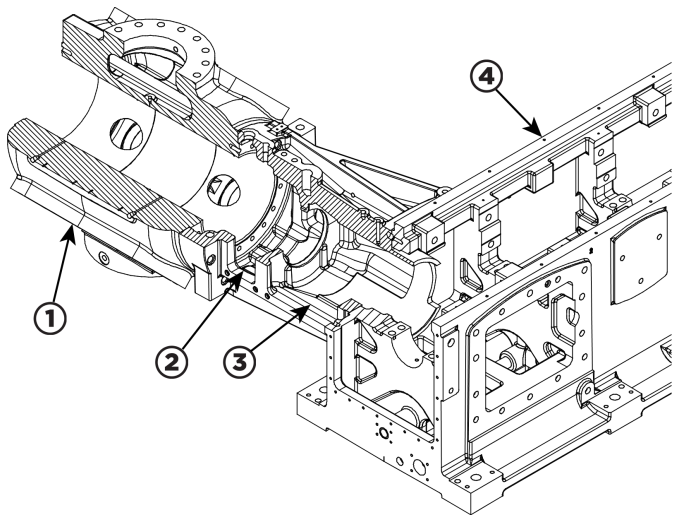
This document pertains to Ariel cylinders and distance pieces preserved at Ariel and/or stored for 12 months since the last preservation. These procedures prevent internal corrosion for 12 months if protection is not compromised. If any deterioration occurs, re-preserve according to the guidelines in this document.

For new compressors to be stored longer than 12 months from the Ariel shipment date, see ER-10.5.1 and apply for Deferred Start-up Status to defer the standard warranty.

Related Documents:

- ER-34 - Preserving Ariel Non-Lube Cylinders and Distance Pieces with VCI Powder
- ER-129 - Preserving Ariel Reciprocating Compressor Frames, Mineral, Synthetic PAO, and Synthetic Ester Oil Compatible
- ER-140 - Preserving Ariel Reciprocating Compressor Frames, PAG Oil Compatible
- ER-141 - Preserving Ariel Cylinders and Distance Pieces, PAG Oil Compatible

Figure: Typical Reciprocating Compressor



1 Cylinder
2 Distance Piece
3 Crosshead Cavity
4 Frame

Preservation Materials and Equipment

> **NOTE:**
>
> Cortec products must be stirred before transferring to a smaller container. After transfer, stir or agitate the container before each use. Keep fluids in sealed containers when not in use.

Use the following or their equivalents:

- Contact rust preventative: Mobil Oil Mobilarma 247 (alternative: Cortec VpCI-322).
- Liquid vapor phase corrosion inhibitor (VCI): Cortec VpCI-329 (alternative: VpCI-322). VpCI-329 and VpCI-322 are equivalent in strength and compatible with each other.
- Protective coating: Cortec VpCI-368 (alternative: marine grade grease).
- Waterproof back, UV resistant tape: Shurtape PC667.
- Shipping covers and gaskets to seal cylinder suction and discharge openings, and standard distance piece covers. Install a standard guide cover in place of any cover equipped with an over-pressure relief valve.
- Wax impregnated cloth: Versil-Pak 3-2 JAN-B-121, Type II Grade C, Class II. Available from Daubert Cromwell LLC (Alsip, Illinois).
- Tarps and tie down materials for outdoor storage.

Preservation Procedure

1. Inspect and preserve in a clean, dry environment. Remove the following:
   - Distance piece guide covers, including any covers equipped with over-pressure relief valves.
   - Suction and discharge cylinder nozzle shipping covers.
   - One crank end suction valve from each cylinder.
   - One head end suction valve from each cylinder.
   - For packaged compressors, remove one discharge valve cap from each cylinder.
2. Inspect all internal surfaces and cavities.
   a. Inspect for corrosion, standing water, sludge, or gel. If found, photograph affected areas and contact Ariel.
   b. Remove water contamination, corrosion, sludge, or gel prior to coating parts with rust preventative.
   c. Inspect for potential leak paths for water and air and seal them.
   d. Verify metal and plastic plugs are tight and replace damaged plugs.
   e. Inspect exposed stainless steel tubing and verify metal tubing caps are tight.
   f. Inspect suction and discharge nozzle shipping covers and gaskets for damage.
3. Thoroughly agitate rust preventative and VCI before each use. Use VCI within 24 months of the manufacture date.
   a. Spray rust preventative inside each compressor cylinder and distance piece. Cover all internal surfaces, including: unloaders, piston rod, valve pocket surfaces, retainers, valve caps, suction and discharge gas passages, valves, and distance pieces.
   b. Allow no noticeable accumulation of rust preventative in the bottom of treated cavities.
   c. For models with top and or side covers with o-ring seals: lightly coat the top rail of the frame and side openings of the guide with protective coating before installing top or side covers.
4. Pour, spray, or fog VpCI corrosion inhibitor into distance piece cavities per Table: Cortec VpCI-329 or 322 Min. Quantity to Preserve Distance Piece. If pouring, distribute evenly throughout. Do not pour VpCI directly on any bearing surfaces.

Figure: Compressor Guide Compartments

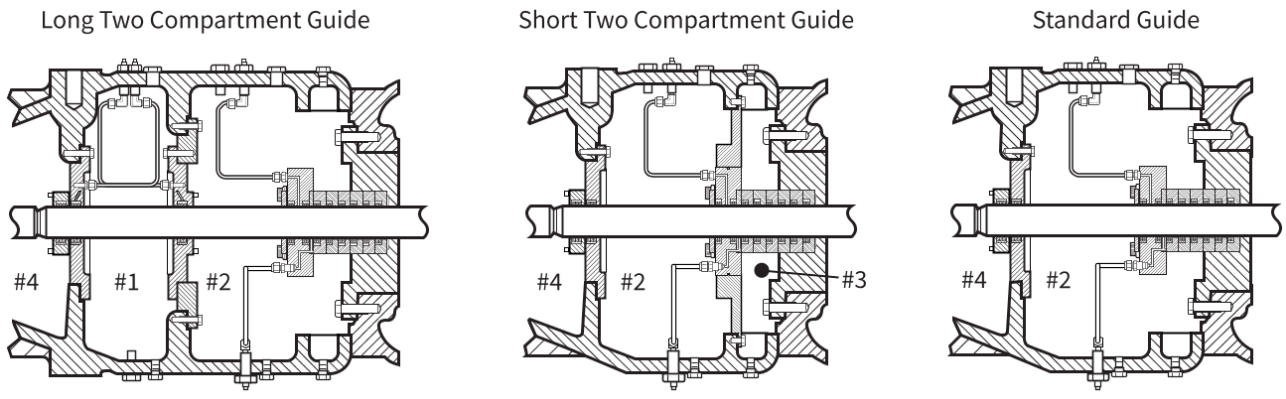


Table: Cortec VpCl-329 or 322 Min. Quantity to Preserve Distance Piece - Fluid Oz. (ml) [a]

Compressor Frame	Cavity #1	Cavity #2	Cavity #3	Cavity #4 [b]
JG:A:R:J	1/2 (15)	1/2 (15)	1/2 (15)	N/A
JGH:E:K:T, KBE:K:T	1/2 (15)	1 (30)	1/2 (15)	2 (59)
JGC:D:F, KBC:D:F	1 (30)	2 (59)	1/2 (15)	4 (118)
JGZ:U, KBZ:U	2 (59)	2 (59)	1 (30)	4 (118)
JGB:V, KBB:V	3 (89)	3 (89)	1-1/2 (44)	6 (177)

a. Refer to Figure: Compressor Guide Compartments
b. Cavity #4 is preserved only when the guide is unmounted with a shipping cover installed on the frame end of the guide. Cavity #4 uses the same preservation method as the frame.

5. Add VCI to each cylinder per Table: Min. Quantity of Cortec VpCl-329 or 322 to Preserve Cylinders. Pour, spray, or fog VCI into the cylinder. If pouring, distribute evenly throughout the cavity. Distribute 75% evenly between the cylinder suction and discharge gas passages, and 25% evenly between the crank end and head end of the cylinder bore. If your compressor does not appear in Table: Min. Quantity of Cortec VpCl-329 or 322 to Preserve Cylinders, contact Ariel for the correct VCI quantities.

Table: Minimum Quantity of Cortec VpCl-329 or 322 to Preserve Each Cylinder

Cylinder Frame Class	Cylinder Bore Diameter - inches						
JG:A:M:P:N:Q:R:J, KB100	< 10	10 to 15	> 15				
JGH:E:K:T, KBE:K:T	< 5			5 to 15	> 15		
JGC:D:F:Z:U, KBC:D:F:Z:U	< 5		5 to 10			> 10	
JGB:V, KBB:V				5 to 10		>10 to 15	> 15
Minimum Quantity per Cylinder, Fluid Oz. (ml)	**1 (30)**	**2 (59)**	**3 (89)**	**4 (118)**	**6 (177)**	**7 (207)**	**9 (266)**
If piston is removed, additional cylinder bore quantity, Fluid Oz. (ml)	**1 (30)**	**1 (30)**	**1 (30)**	**2 (59)**	**3 (89)**	**4 (118)**	**5 (148)**

6. VCI requires protected cavities to remain sealed during storage. Open cavities deplete VpCI concentration, rendering it ineffective. To prevent VCI depletion, seal the compressor as follows:
   - Plug all openings, including vents and breather connections. Verify plugs are tight.
   - Reassemble all covers and valve assemblies.
   - Seal cylinder suction and discharge nozzles with shipping covers and gaskets.
   - Verify proper installation of cylinder shipping covers and gaskets.
   - Replace damaged shipping flange covers or gaskets.
   - Hand-wrench tighten shipping flange cover bolts in a criss-cross pattern.
7. Tag distance piece/cylinder assembly with the date of the last inspection and preservation.

Storage

Prior to packaging, store in a clean, dry environment. If stored outdoors, protect from direct contact with the elements. Never store with the mounting feet in direct contact with the ground; protect them from corrosion. If stored after packaging, protect all piping and bottles in the package as well. Verify all openings in the package are sealed.

Commissioning Compressor to Service

1. Inspect all compressor internal surfaces and cavities. Inspect for corrosion, standing water, sludge, or gel. If found, photograph affected areas and contact Ariel.
2. Change the oil filter. There is no need to change oil, or if drained, to flush the frame prior to filling with oil; rust preventative and VCI will mix with the lube oil. For PAG synthetic oil applications, consult Ariel.
3. Remove temporary covers, gaskets, plugs, tape, tags, etc. Replace all connections, including vents and drains. The installation guide covers equipment equipped with relief valves, if applicable.
4. Complete the Start-Up Check list (ER-10.6).

CHANGE DESCRIPTION: 1st ¶: inserted 'preserved at Ariel and/or'.